a Resolution ladder, deepest cell

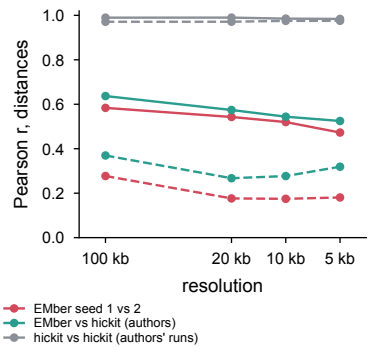
GM12878 scMicro-C cell 035 (GSM5764669), 3.01 M contacts; EMber diploid, frozen defaults, one H100; r: Pearson, distances of random particle pairs

final resolution	particles (M) grid / simulated	contacts per particle	models	wall time (min)	peak GPU memory (GB)	seed 1 vs 2, r per copy / whole	EMber vs hickit, r per copy / whole	hickit vs hickit, r per copy / whole
100 kb	0.06 / 0.06	51.74	10	1.4	0.5	0.58 / 0.28	0.64 / 0.37	0.99 / 0.97
20 kb	0.29 / 0.27	10.36	10	2.2	1.2	0.54 / 0.18	0.57 / 0.27	0.99 / 0.97
10 kb	0.58 / 0.53	5.18	10	3.1	1.7	0.52 / 0.17	0.54 / 0.28	0.99 / 0.98
5 kb	1.16 / 1.05	2.59	10	4.6	2.8	0.47 / 0.18	0.52 / 0.32	0.98 / 0.98
2 kb *	2.90 / 2.34	1.04	5	4.6	3.0	– / –	– / –	– / –
1 kb *	5.80 / 3.60	0.52	5	6.7	3.6	– / –	– / –	– / –

* exploratory (5 models, seed 1). Wall time: one process, contacts read to models written. Simulated particles: those holding a contact end (adaptive resolution below 25 kb; the others interpolated). Per copy chain: median over the 46 chains. hickit: the authors' five runs (5 / 10 / 20 kb; at 100 kb their 20 kb runs on the 100 kb grid), every EMber model vs every run.

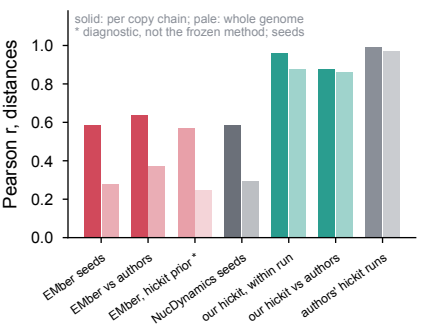
b Agreement vs resolution

cell 035; per copy chain (solid), whole genome (dashed)



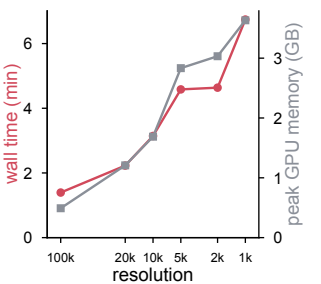
c Controls at 100 kb

cell 035; models of one run vs another (or vs the authors')



d Time and GPU memory

cell 035, one H100



e Five deepest cells, 5 kb

10 models; seeds 1 and 2; r per copy chain

cell	contacts (M)	wall (min)	GPU (GB)	EMber seeds	EMber vs hickit	hickit runs
035	3.01	4.6	2.8	0.47	0.52	0.98
024	2.85	4.7	2.8	0.50	0.55	0.99
320	2.64	4.9	2.7	–	–	0.76
047	2.24	4.4	2.7	–	–	–
028	2.23	4.2	2.7	–	–	–

Held-out contacts at 10 kb (cell 035; 10 % of the reads held out, one fold; separation-matched never-contacted negatives): strict: cis ROC-AUC 0.66, trans 0.60; cis < 2 Mb same homolog: cis ROC-AUC 0.66, trans 0.57.

On this data the native engine's diploid formulation (EMber and the NucDynamics 2017 protocol, every unphased contact one ambiguous restraint over its candidate copy pairs) does not reproduce between runs, at any resolution and separation, while hickit's pipeline (2-D phase imputation, homologs split at p >= 0.75, isolated-contact filters -c1 / -c5) does; on Dip-C GM12878 (1 M contacts) EMber reached r 0.98 between seeds and 0.90 vs the published hickit structure with the same metric. EMber's own phase prior is not the cause (hickit's prior: same agreement). Its noise estimate is bistable here (eps 0.5 in the 100 kb runs, 1e-4 in the others). Not tuned (demonstration).